



Mathematics Written Assessment 2024 Elementary District Governor's Cup

Student Instructions:

- For each question, choose the BEST answer.
1. Which of these numbers is a natural number?
A. 0
B. -23
C. $\frac{1}{5}$
D. 19
 2. What is the value of 28^2 ?
A. 78,400
B. 7,840
C. 784
D. 78.4
 3. The temperature at noon was 54° . By 4 PM, the temperature had risen 8° and then by 7 PM, the temperature had dropped 18° . What was the temperature at 7 PM in degrees?
A. 44°
B. 83°
C. 36°
D. 52°
 4. How many even prime numbers are there less than 100?
A. 0
B. 1
C. 12
D. 25
 5. Arrange the following fractions in ascending order: $\frac{1}{16}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{25}$
A. $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{25}$
B. $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{4}$, $\frac{1}{25}$
C. $\frac{1}{16}$, $\frac{1}{8}$, $\frac{1}{25}$, $\frac{1}{4}$
D. $\frac{1}{25}$, $\frac{1}{16}$, $\frac{1}{8}$, $\frac{1}{4}$
 6. Which of the following data is an example of primary quantitative data?
A. The program book in a basketball game shows the pictures of individual players.
B. You count the number of points scored by each individual player in a basketball game.
C. You describe the colors of the basketball players' uniforms.
D. You observe that some members of the team wore a different style of shoes.

7. A data set has the elements {26, 43, 52, 96, 100}. Find the median of the data set.
- A. 43
 - B. 47.5
 - C. 52
 - D. 96

Use this tally chart to answer questions 8 – 10.

Favorite Governor's Cup Competitions	Number of Competitors
Math Content Assessment	
Science Content Assessment	
Social Studies Content Assessment	
Language Arts Content Assessment	
Arts & Humanities Content Assessment	
Composition	
Future Problem Solving	
Quick Recall	

8. How many competitors are shown in the tally chart?
- A. 42
 - B. 43
 - C. 44
 - D. 49
9. Based on the information in the tally chart, which competition is the least favorite?
- A. Science Content Assessment
 - B. Social Studies Content Assessment
 - C. Composition
 - D. Quick Recall
10. How many more competitors prefer content assessments to Quick Recall?
- A. 24
 - B. 12
 - C. 5
 - D. Impossible to tell from the data
11. Find all the prime factors of 75.
- A. 1, 75
 - B. 3, 25
 - C. 5, 3
 - D. 5, 15

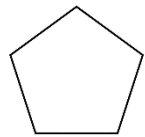
12. Aneesh wanted to go to the carnival. Admission tickets are \$15.25 which includes five games. Any additional games are \$1.25 each. If Aneesh plays ten games, how much money will he need to pay for his trip to the carnival?
- A. \$21.50
 - B. \$16.50
 - C. \$12.50
 - D. \$6.25
13. Which fraction is equal to $\frac{2}{3}$?
- A. $\frac{4}{5}$
 - B. $\frac{8}{9}$
 - C. $\frac{6}{9}$
 - D. $\frac{6}{12}$
14. Happy the Hippo loves flowers. He bought a bouquet of two dozen roses. One-half of the roses were red and $\frac{1}{3}$ of the roses were yellow. The rest of the roses were white. How many roses were white?
- A. 12
 - B. 3
 - C. 6
 - D. 4
15. The recipe for banana pudding calls for $\frac{3}{4}$ cup of condensed milk. If you need to triple the recipe, how many cups of condensed milk will you need?
- A. 3
 - B. $2\frac{1}{4}$
 - C. $3\frac{3}{4}$
 - D. $2\frac{1}{2}$
16. The ratio of girls to boys in Mrs. Thompson's class is 2:1. If there are 30 students in her class, how many girls and how many boys are there in the class?
- A. 20 boys and 10 girls
 - B. 20 girls and 10 boys
 - C. 28 girls and 2 boys
 - D. 2 girls and 28 boys
17. A survey showed that there are 22 oak trees in the park and 11 willow trees. What is the simplified ratio of the number of oak trees to the total number of trees in the park?
- A. 2:3
 - B. 22:33
 - C. 2:1
 - D. 22:11

18. It takes Joey 20 minutes to ride his bike to school. The school is 2 miles from his home. The park is 3 miles from Joey's house. If he rides to the park at the same speed, how long will it take him to get to the park?
- A. $\frac{1}{3}$ hour
 - B. 25 minutes
 - C. 40 minutes
 - D. 30 minutes
19. Coats are on sale at the mall for 30% off the regular price. The regular price for a coat is \$55.00. What will a customer pay if he buys 2 coats during the sale, ignore any tax?
- A. \$16.50
 - B. \$32.50
 - C. \$77.00
 - D. \$77.50
20. There are 5,280 feet in one mile. How many inches are in one mile?
- A. 63,360
 - B. 440
 - C. 1,760
 - D. 63,380
21. Which is more, 1 yard or 48 inches? How much more?
- A. 1 yard, 36 inches more
 - B. 48 inches, 1 foot more
 - C. 1 yard, 2 inches more
 - D. 48 inches, 2 feet more.
22. Carol needs 5 meters of ribbon to make a large wreath for the stage. Each spool of ribbon is 200 centimeters long. How many spools will Carol need to buy to make sure she has enough ribbon?
- A. 5
 - B. 4
 - C. 3
 - D. 2
23. Begin at the origin of the coordinate plane. Move up 4 and to the right 3, what is the ordered pair showing the point's location?
- A. (4, -3)
 - B. (-4, 3)
 - C. (3, -4)
 - D. (3, 4)
24. Line AB is perpendicular to Line CD. What is the measure of each of the four angles formed at the intersection of the two lines?
- A. There is not enough information to determine the measure of the angles.
 - B. The lines do not intersect, so there are no angles formed.
 - C. Each angle measures 90 degrees.
 - D. Each angle measures 360 degrees.

25. One angle of a right triangle measures 45 degrees. What are the measures of the other two angles?
- A. 90 degrees, 30 degrees
 - B. 90 degrees, 45 degrees
 - C. 45 degrees, 45 degrees
 - D. 90 degrees, 35 degrees

26. What is the name of an angle that measures 91 degrees?
- A. Acute angle
 - B. Obtuse angle
 - C. Right angle
 - D. Straight angle

27. Look at the regular polygon with five sides on the right. What is the name of the shape and how many degrees are in each of the interior angles of the shape?
- A. Pentagon, 108°
 - B. Pentagon, 140°
 - C. Pentagon, 360°
 - D. Pentagon, 540°



28. How many diagonals are in a regular hexagon?
- A. 2
 - B. 5
 - C. 6
 - D. 9
29. Sally wanted to fence in her flower garden. Her garden is shaped like a regular heptagon. Each side of the garden measures 8 feet. How many feet of fence will she need?
- A. 40
 - B. 48
 - C. 56
 - D. 64
30. Porky's Pizza sells two different sized pizzas. They have five different toppings and two different types of cheese listed on the menu. How many different pizzas can Porky's Pizza make, using only one topping and 1 type of cheese?
- A. 20
 - B. 10
 - C. 9
 - D. 4

The next 6 questions are Tiebreakers.

31. The game of Happy Times has a spinner divided into 8 equal sectors – 2 blue sectors, 4 red sectors, 1 green sector, and 1 orange sector. To win, Hailee needs to land on a blue sector. What is the probability that the spinner will land on blue?
- A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
 - C. $\frac{1}{8}$
 - D. $\frac{2}{4}$
32. What is the value of 7 cubed minus 4 squared?
- A. 0
 - B. 3
 - C. 327
 - D. 343
33. Solve the following equation for x: $x + 74 = 2x + 3$.
- A. 71
 - B. -71
 - C. 75
 - D. -75
34. Triangular numbers are numbers shown as dots that form equilateral triangles. What is the 8th triangular number?
- A. 28
 - B. 36
 - C. 45
 - D. 55
35. Identify the exponent in the following polynomial. $27x^5 + 3xy + 9y$
- A. 27
 - B. 5
 - C. 3
 - D. 9
36. Expand the following algebraic expression and combine like terms. $9(x + 4) - 3(x - 8)$
- A. $9x + 36 - 3x - 24$
 - B. $9x + 36 + 3x + 24$
 - C. $6x + 60$
 - D. $6x - 60$

Elementary – Mathematics – District
2024 Governor’s Cup – KEY

1. D
2. C
3. A
4. B
5. D
6. B
7. C
8. D
9. B
10. B
11. C
12. A
13. C
14. D
15. B
16. B
17. A
18. D
19. C
20. A
21. B
22. C
23. D
24. C
25. B
26. B
27. A
28. D
29. C
30. A

TIE BREAKERS

31. B
32. C
33. A
34. B
35. B
36. C